

42112 — Memorable Stories Cheat Sheet

§6.5 Micro Brewery 2

The story: A brewer can only run one beer through the tank per month. Everything you don't sell sits in storage costing money.

Element	Story
Demand [b,m]	Thirsty customers — how many litres they want this month
BrewCap = 120	The tank holds 120 litres max
StoreCap = 300	The cellar holds 300 litres total
Cost = 0.1	Every litre sitting in the cellar costs 10 cents a month
InitialStorage [b]	What's already in the cellar when you start January
x [b,m]	How much you brew this month
y [b,m]	How full the cellar is at the end of the month
q [b,m]	Did the brewery run this month? Yes or no
Objective	Minimise the cellar bill — every litre stored costs money
Inventory balance	Cellar tonight = cellar last night + what you brewed – what you sold
$x \leq \text{BrewCap} \cdot q$	You can't brew anything if the brewery isn't running, and if it is, max 120L
$q \leq 1$	Only one tap on the tank — one beer type at a time
$y \leq \text{StoreCap}$	The cellar only fits 300 litres total

§6.9 TA Workplan

The story: Four teaching assistants need to staff a help desk. Students need help at specific times. TAs hate inconvenient shifts. Nobody should arrive, leave, come back.

Element	Story
Demand [p,d]	How many TAs must be at the desk at this hour
Inconvenience [ta,p,d]	How much this particular TA hates this particular slot
x [ta,p,d]	Is this TA standing at the desk right now?
y [ta,p,d]	Did this TA just walk in the door this slot?
Objective	Minimise total TA misery
x Demand	Enough bodies at the desk every hour
x = 52	Each TA works their contracted hours, no more, no less
y 1	You can only arrive once per day
x x _{prev} + y	You're only here if you were here last hour, or you just arrived
x 2 · y	If you bothered coming in, you stay at least 2 hours

§6.12 Tennis

The story: A court sweeper must drag a brush over every line. They can walk between lines freely but want to minimise total footsteps.

Element	Story
Distance [p1,p2]	How far apart two intersection points are
Lines [p1,p2]	Is there a line between these two points that needs sweeping?
x [p1,p2]	Does the sweeper walk from point A to point B?
s [p]	Did the sweeper start their shift here?
e [p]	Did the sweeper finish their shift here?
Objective	Minimise total steps taken
Flow conservation	Every time you walk into a point, you also walk out — you can't teleport
s = 1, e = 1	One place to start, one place to finish
x [p1,p2] + x [p2,p1] Lines	Every line gets swept — at least one direction

Element	Story
fix(s/e = 0 for interior)	You can't start or end in the middle of the court — you'd have to walk back across clean clay

§6.14 Tariff Rates

The story: A power company decides each night how many generators to run across five time blocks. Starting a generator costs a one-time fee. The plan repeats every day, so yesterday's last shift is today's first shift's predecessor.

Element	Story
PDEM[h]	How hungry the grid is in this time block
NHP[h]	How many hours this block lasts
NG[p]	How many generators of this type the company owns
MINL/MAXL[p]	Every generator has a minimum it must run at, and a ceiling it can't exceed
MINC[p]	What it costs per hour just to keep one generator ticking over at minimum
PC[p]	Extra cost per MW above minimum — ramping up is expensive
STARTC[p]	The ignition fee — every time a cold generator starts
nRun[p,h]	How many of this generator type are humming right now
prod[p,h]	Total MW this generator type is actually pushing to the grid
nStart[p,h]	How many generators of this type just switched on this block
Objective	Pay the running bill + the ramping bill + the ignition bill
prod MINL · nRun	A running generator can't idle below its minimum — physics
prod MAXL · nRun	A running generator can't exceed its rated maximum — also physics
prod PDEM	The lights stay on — demand is always met

Element	Story
$nStart \quad nRun - nRun_prev$	If more are running now than last block, someone must have pressed the start button
$MAXL \cdot nRun \quad 1.15 \cdot PDEM$	Even if demand spikes 15%, enough generators are warm to cover it

§6.17 Wedding Planner

The story: A wedding planner seats 20 guests across 3 tables. Couples must stay together. Beyond that, the planner wants guests who share interests close together, genders balanced, and nobody surrounded by strangers.

Element	Story
$TableCap = 9$	No table can be crowded beyond 9 seats
$Couple[g1,g2]$	These two come as a package deal
$SharedInterests[g1,g2]$	How much these two have to talk about
$Male[g] / Female[g]$	Guest g's gender flag
$Know[g1,g2]$	Do these two already know each other?
$x[g,t]$	Is this guest sitting at this table?
$y[g1,g2,t]$	Are these two guests sitting at the same table?
$m[t] / f[t]$	How many too many men / women are at this table beyond the allowed imbalance of 2
$k[g]$	How many familiar faces this guest is missing — shortfall below 3 known people
Objective 17.2	Pack each table with people who have things in common
Objective 17.3	Minimise gender imbalance across tables
Objective 17.4	Minimise the number of guests who feel like strangers
Objective 17.5	Do all three at once
$x = 1$	Every guest sits somewhere — exactly once
$x \leq TableCap$	Don't overcrowd the table
$x[g1,t] = x[g2,t]$	Couples are glued together

Element	Story
$y \leq x[g1,t]; y \leq x[g2,t]$	You can only count a pair as “together” if both are actually at that table
$\text{Male} \cdot x - \text{Female} \cdot x \leq 2 + m[t]$	At most 2 more men than women — or pay a penalty
$k - 3 \cdot x + \text{Know} \cdot y \geq 0$	At your table you should know at least 3 people — or the shortfall is counted against you

SIP Production Planning

The story: A 3D printing shop picks which customer orders to run today. Three machines all have 8-hour limits. In Assignment 2, delivering to a buyer costs a flat fee whether you send them one item or ten.

Element	Story
Price[i]	What the customer offered to pay
PrintTime / PolishTime / PaintTime[i]	How long each machine needs for this item
Minutes = 480	Each machine runs for 8 hours — then it stops
BuyerId[i]	Whose order is this item part of?
TransportCost = 50	One delivery van per buyer, regardless of how many items
$x[i]$	Do we produce this item today?
$y[b]$	Are we sending a delivery to this buyer today?
Objective 1	Maximise today’s revenue
Objective 2	Maximise revenue minus delivery costs
PrintTime · $x \leq 480$	The printer can’t run beyond its shift
PolishTime · $x \leq 480$	The polisher can’t run beyond its shift
PaintTime · $x \leq 480$	The painter can’t run beyond its shift
$x[i] \leq y[\text{BuyerId}[i]]$	You can’t produce an item without sending a delivery to its buyer